

Cisco NX-OS System Management Configuration Guide

Section 1: Overview of System Management Features

Cisco NX-OS System Management features provide enterprise control for configuring, monitoring, and maintaining network devices. Key features include: - Secure Shell (SSH) and Telnet management protocols - Network Time Protocol (NTP) time synchronization across distributed data center switches - Simple Network Management Protocol (SNMP v2c/v3) monitoring and trap generation - Web-based management access via HTTP and HTTPS (Cisco NX-API) - System Logging (Syslog) and telemetry streaming - AAA (Authentication, Authorization, and Accounting) via RADIUS and TACACS+

Section 2: Configuring SSH (Secure Shell)

To configure SSH on Cisco NX-OS switches, perform the following steps: 1. Enable the SSH feature: `feature ssh` 2. Generate RSA host key pair: `crypto key generate rsa force label ssh-key 2048` 3. Restrict SSH to protocol version 2: `ip ssh version 2` 4. Verify SSH server status: `show ssh server` Note: Disabling Telnet (`no feature telnet`) is mandatory for enterprise compliance.

Section 3: Configuring and Verifying NTP Synchronization

To ensure consistent log timestamps across all network devices, configure NTP: 1. Configure primary NTP server: `ntp server 192.168.1.100 prefer use-vrf management` 2. Specify NTP source interface: `ntp source-interface mgmt0` 3. Enable NTP authentication if required: `ntp authentication-key 1 md5 Cisco123` 4. Verify NTP synchronization status: `show ntp peer-status` 5. Verify detailed clock synchronization: `show ntp status` Output must display 'NTP server synchronized' and stratum level 2 or lower.

Section 4: Disabling Unsecure HTTP and HTTPS Access

For enterprise security hardening, unsecure Web / HTTP management must be disabled: 1. Disable HTTP server feature: `no feature http` 2. Disable unsecure HTTP server: `no ip http server` 3. Disable secure HTTPS web management (if required): `no ip http secure-server` 4. Disable NX-API web server: `no feature nxapi` 5. Verify feature state: `show feature | include http` Verification Output must confirm HTTP feature is disabled ('disabled').